



Experiment No.: 3

Title: Implement control structures and form validation; create a grading system based on user-entered marks.

Software/Tools Required: HTML5, CSS3, JavaScript, VS Code , Google Chrome

Index.html:

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Student Grading System</title>
  <link rel="stylesheet" href="style.css">
</head>
<body>
  <div class="container">
    <h1>Student Grading System</h1>
    <form id="gradeForm">
      <div class="input-group">
        <label>Student Name</label>
        <input type="text" id="studentName" placeholder="Enter Student Name">
      </div>
      <div class="input-group">
        <label>Roll Number</label>
        <input type="text" id="rollNumber" placeholder="Enter Roll Number">
      </div>
      <div class="input-group">
        <label>Javascript</label>
        <input type="number" id="javascript" min="0" max="100">
      </div>
      <div class="input-group">
        <label>Computer Networks</label>
        <input type="number" id="computer-networks" min="0" max="100">
      </div>
      <div class="input-group">
        <label>Compiler Construction</label>
        <input type="number" id="compiler-construction" min="0" max="100">
      </div>
      <div class="input-group">
        <label>Data Compression</label>
        <input type="number" id="data-compression" min="0" max="100">
      </div>
    </form>
  </div>
</body>
</html>
```



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```
</div>
<div class="input-group">
  <label>Specialization(Data Science/AI/ML)</label>
  <input type="number" id="specialization" min="0" max="100">
</div>
<button type="submit">Calculate Result</button>
</form>
<div id="result"></div>
</div>
<script src="script.js"></script>
</body>
</html>
```

Style.css:

```
*{
  margin:0;
  padding:0;
  box-sizing:border-box;
  font-family: Ballance;
}
body{
  background:linear-gradient(135deg,#99c6ee,#a4e5f0);
  display:flex;
  justify-content:center;
  align-items:center;
  min-height:100vh;
}

.container{
  width: 1000px;
  background:#c7e0f4;
  padding:30px;
  border-radius:15px;
  box-shadow:0 10px 25px rgba(0,0,0,.2);
}
h1{
  text-align:center;
  margin-bottom:25px;
  color:#0e0e0f;
}
.input-group{
  margin-bottom:15px;
}
label{
```



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```
display:block;
margin-bottom:6px;
font-weight:bold;
}
input{
width:100%;
padding:10px;
border:1px solid #d2e4fd;
border-radius:8px;
font-size:15px;
}
input:focus{
outline:none;
border-color:#0d6efd;
}
button{
width:100%;
padding:12px;
background:#0c48a1;
color:white;
border:none;
border-radius:8px;
font-size:16px;
cursor:pointer;
}
button:hover{
background:#084298;
}
#result{
margin-top:20px;
display:none;
padding:15px;
background:#f5f8ff;
border-radius:10px;
}
table{
width:100%;
border-collapse:collapse;
}
td{
padding:10px;
border-bottom:1px solid #ddd;
}
.success{
```



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```
color:green;
font-weight:bold;
}
.fail{
color:red;
font-weight:bold;
}
```

Script.js:

```
const gradeForm = document.getElementById("gradeForm");
gradeForm.addEventListener("submit", function (event) {
    event.preventDefault();
    const name = document.getElementById("studentName").value.trim();
    const rollNumber = document.getElementById("rollNumber").value.trim();
    const javascript = Number(document.getElementById("javascript").value);
    const computerNetworks = Number(document.getElementById("computer-
networks").value);
    const compilerConstruction = Number(document.getElementById("compiler-
construction").value);
    const dataCompression = Number(document.getElementById("data-compression").value);
    const specialization = Number(document.getElementById("specialization").value);
    const marks = [
        javascript,
        computerNetworks,
        compilerConstruction,
        dataCompression,
        specialization
    ];
    if (name === "") {
        alert("Please enter Student Name.");
        return;
    }
    if (rollNumber === "") {
        alert("Please enter Roll Number.");
        return;
    }
    for (let mark of marks) {
        if (isNaN(mark)) {
            alert("Please enter all subject marks.");
            return;
        }
        if (mark < 0 || mark > 100) {
            alert("Marks should be between 0 and 100.");
            return;
        }
    }
}
```



```
}
const total = marks.reduce((sum, mark) => sum + mark, 0);
const percentage = (total / 500) * 100;
let status = "Pass";
for (let mark of marks) {
  if (mark < 35) {
    status = "Fail";
    break;
  }
}
let grade;
if (status === "Fail") {
  grade = "F";
} else if (percentage >= 90) {
  grade = "A+";
} else if (percentage >= 80) {
  grade = "A";
} else if (percentage >= 70) {
  grade = "B";
} else if (percentage >= 60) {
  grade = "C";
} else if (percentage >= 50) {
  grade = "D";
} else {
  grade = "F";
}
let remarks;
switch (grade) {
  case "A+":
    remarks = "Outstanding";
    break;
  case "A":
    remarks = "Excellent";
    break;
  case "B":
    remarks = "Very Good";
    break;
  case "C":
    remarks = "Good";
    break;
  case "D":
    remarks = "Needs Improvement";
    break;
  default:
```



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```
        remarks = "Better Luck Next Time";
    }
    const studentResult = {
        name,
        rollNumber,
        total,
        percentage: percentage.toFixed(2),
        grade,
        status,
        remarks
    };
    localStorage.setItem("studentResult", JSON.stringify(studentResult));
    displayResult(studentResult);
});
function displayResult(data) {
    const result = document.getElementById("result");
    result.style.display = "block";
    result.innerHTML = `
        <h2 style="text-align:center; margin-bottom:15px;">
            Student Result
        </h2>
        <table>
            <tr>
                <td><strong>Student Name</strong></td>
                <td>${data.name}</td>
            </tr>
            <tr>
                <td><strong>Roll Number</strong></td>
                <td>${data.rollNumber}</td>
            </tr>
            <tr>
                <td><strong>Total Marks</strong></td>
                <td>${data.total} / 500</td>
            </tr>
            <tr>
                <td><strong>Percentage</strong></td>
                <td>${data.percentage}%</td>
            </tr>
            <tr>
                <td><strong>Grade</strong></td>
                <td>${data.grade}</td>
            </tr>
            <tr>
                <td><strong>Status</strong></td>
                <td>${data.status}</td>
            </tr>
        </table>
    `;
}
```



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Experiment-3

```
<td class="${data.status === "Pass" ? "success" : "fail"}">
    ${data.status}
</td>
</tr>
<tr>
<td><strong>Remarks</strong></td>
<td>${data.remarks}</td>
</tr>
</table>
`;
}
window.onload = function () {
    const savedData = localStorage.getItem("studentResult");
    if (savedData) {
        displayResult(JSON.parse(savedData));
    }
};
```

The screenshot shows a web browser window with the title "Student Grading System". The address bar shows the file path: C:/Users/hp/Documents/24070521148_ShrutiBawankar_C1/JavaScript_Lab/Experiments/EXP3/index.html. The browser has tabs for "Student Grading System" and "Ask Gemini". The page content is a form with a light blue background. The form has the following fields and values:

Field	Value
Student Name	Shruti Bawankar
Roll Number	24070521148
Javascript	88
Computer Networks	78
Compiler Construction	86
Data Compression	84
Specialization(Data Science/AI/ML)	90

At the bottom of the form is a blue button labeled "Calculate Result".



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Experiment-3

Student Grading System

File C:/Users/hp/Documents/24070521148_ShrutiBawankar_C1/JavaScript_Lab/Experiments/EXP3/index.html

Specialization(Data Science/AI/ML)

90

Calculate Result

Student Result

Student Name	Shruti Bawankar
Roll Number	24070521148
Total Marks	426 / 500
Percentage	85.20%
Grade	A
Status	Pass
Remarks	Excellent

Result: The Student Grading System was successfully implemented using HTML, CSS, and JavaScript. It validates user input, calculates the percentage, assigns grades using control structures, and displays the result accurately on a separate result page.

Conclusion: The Student Grading System successfully demonstrates the use of **HTML, CSS, and JavaScript** to create an interactive web application. It implements **form validation, control structures, loops, and Local Storage** to process user input and generate an accurate student result. The project provides a professional and user-friendly interface while reinforcing fundamental concepts of client-side web development.